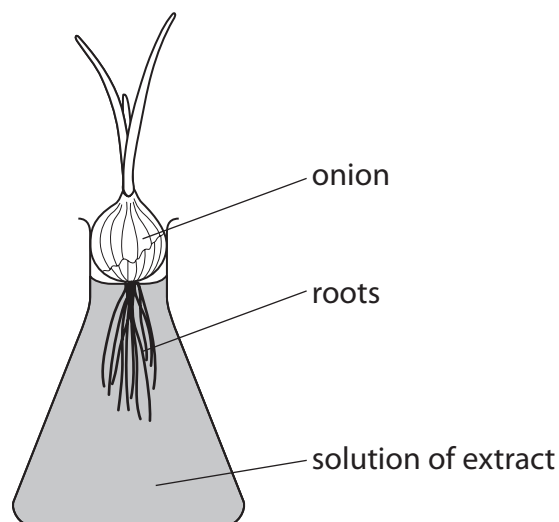


- 2 *Psychotria* species are rainforest plants that are widely used as herbal medicines. It has been suggested that extracts from these plants could be used to inhibit cell division.

The effect of extracts from two species of this plant on the mitotic index of onion roots was investigated.

Two concentrations of extract, 1.0 mg cm^{-3} and 7.0 mg cm^{-3} , were made from each of two species of *Psychotria*, species A and species B.

Onion roots were grown in each concentration of extract and a control, as shown in the diagram.



After 24 hours the mitotic index was determined for these roots.

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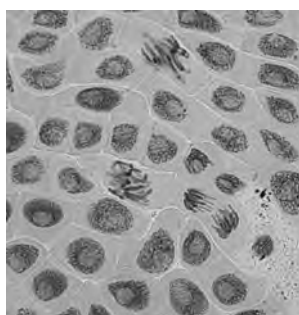
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(a) Describe a safe method to observe the stages of mitosis in these roots.

(5)

(b) The photograph shows part of a root tip from this investigation.



(Source: © Rattiya Thongdumhyu/shutterstock)

Calculate the mitotic index of this root tip.

Give your answer to 2 significant figures.

(2)

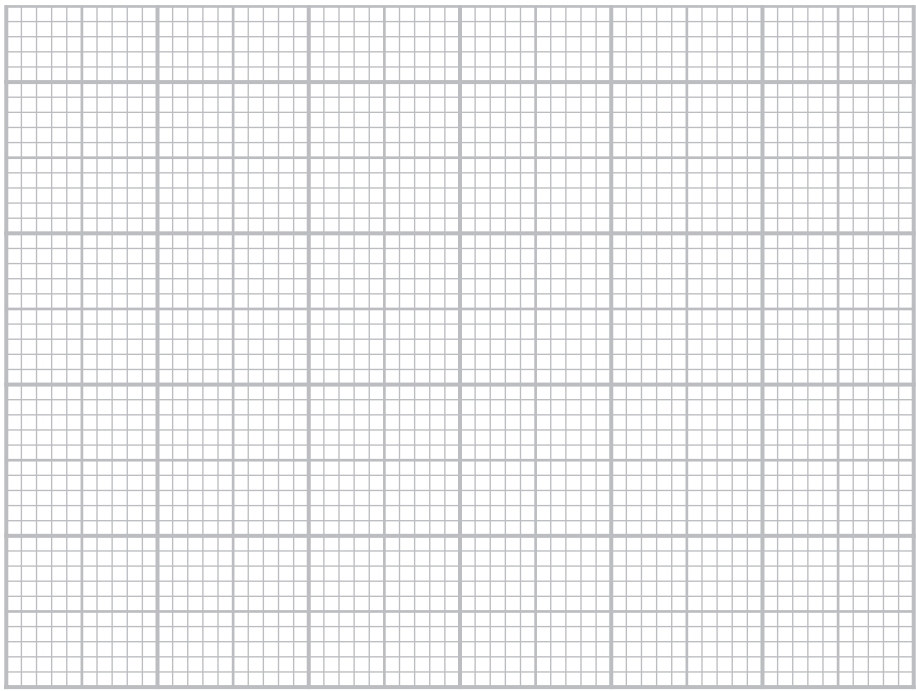
Answer

(c) The table shows data from this investigation.

Source of extract solution	Extract concentration / mg cm^{-3}	Mitotic index
Species A	1.0	4.7
	7.0	1.2
Species B	1.0	4.2
	7.0	1.9
Control	0.0	7.4

(i) Plot a suitable graph to show the data in the table.

(4)



- (ii) Describe how this investigation could be modified to determine if the differences between the mitotic indices are significant.

(4)

(Total for Question 2 = 15 marks)

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